Analogue electronics: mixed questions

National 5 · Electronics & control · Class worksheet

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ choose the right formula for each circuit
- ☐ calculate resistance, current and voltage in series, parallel and combined circuits
- ☐ convert between kΩ and Ω, and A and mA
- ☐ show working and give units

I DO — Key ideas

Show all working and use the correct units

V = IR

Voltage = current × resistance (Ohm's law)

- V in volts, I in amps, R in ohms (Ω)
- Convert first: 4.7 kΩ = 4700 Ω, 30 mA = 0.03 A

	Series	Parallel
Resistance	$R_T = R_1 + R_2 + \ldots$	$1/R_T = 1/R_1 + 1/R_2 + \ldots$ (two: $R_1R_2 \div (R_1 + R_2)$)
Current	The same everywhere	Splits: $I_T = I_1 + I_2 + \ldots$
Voltage	Shared: $V_s = V_1 + V_2 + \ldots$	The same across every branch

YOU DO — On your own

Show your working. Check your answers afterwards.

1 A 12 V supply is connected to a 240 Ω resistor. Calculate the current.

2 15 Ω and 35 Ω resistors are in series on a 10 V supply.

a) Calculate the total resistance.

b) Calculate the current.

c) Calculate the voltage across each resistor.

3 100 Ω and 300 Ω resistors are in parallel on a 12 V supply.

a) Calculate the total resistance.

b) Calculate the total current.

c) Calculate the current through the 100 Ω resistor.

4 A current of 0.5 A flows through a resistor on a 9 V supply. Calculate the resistance.

5 A 20 Ω resistor is in series with 60 Ω and 30 Ω resistors in parallel. The supply is 18 V.

a) Calculate the total resistance.

b) Calculate the current.

c) Calculate the voltage across the 20 Ω resistor.

6 R_1 = 2.2 kΩ and R_2 = 4.7 kΩ are in series on a 12 V supply.

a) Calculate the total resistance.

b) Calculate the current.

c) Calculate the voltage across R_2.

7 10 Ω, 20 Ω and 40 Ω resistors are in parallel on a 12 V supply.

a) Calculate the total resistance.

b) Calculate the total current.

c) Calculate the current through the 20 Ω resistor.

8 A current of 2 A flows through a 15 Ω resistor. Calculate the supply voltage.

9 100 Ω, 150 Ω and 250 Ω resistors are in series on a 20 V supply.

a) Calculate the total resistance.

b) Calculate the current.

c) Calculate the voltage across the 250 Ω resistor.

10 8 Ω and 12 Ω resistors are in parallel on a 6 V supply.

a) Calculate the total resistance.

b) Calculate the total current.

11 A 50 Ω resistor is in series with two 100 Ω resistors in parallel. The supply is 10 V.

a) Calculate the total resistance.

b) Calculate the current.

c) Calculate the current through one of the 100 Ω resistors.

12 A student needs 25 Ω but only has two 50 Ω resistors.

a) Draw a circuit to give 25 Ω.

b) Prove your answer with a calculation.

13 A 24 V supply is connected to a 3 kΩ resistor. Calculate the current in mA.

14 1 kΩ and 2 kΩ resistors are in parallel on a 9 V supply.

a) Calculate the total resistance.

b) Calculate the total current.

15 30 Ω and 60 Ω resistors in parallel are in series with a 10 Ω resistor on a 12 V supply.

a) Calculate the total resistance.

b) Calculate the current.

c) Calculate the voltage across the 10 Ω resistor.

d) Calculate the current through the 30 Ω resistor.

16 R_1 = 470 Ω and R_2 = 1.5 kΩ are in series on a 9 V supply.

a) Calculate the total resistance.

b) Calculate the current.

17 Two heaters (20 Ω and 30 Ω) are in parallel on a 12 V battery.

a) Calculate the total resistance.

b) Calculate the total current.

c) Calculate the current through the 20 Ω heater.

18 A circuit has a total resistance of 40 Ω: a 10 Ω resistor in series with a parallel section. Calculate the resistance of the parallel section.

19 2 kΩ, 5 kΩ and 10 kΩ resistors are in parallel on a 10 V supply.

a) Calculate the total resistance.

b) Calculate the total current.

c) Calculate the current through the 2 kΩ resistor.

20 A 6 V supply feeds a 15 Ω resistor in series with two 30 Ω resistors in parallel.

a) Calculate the total resistance.

b) Calculate the current.

c) Calculate the voltage across the parallel section.

d) Calculate the current through one 30 Ω resistor.

Check and reflect

Mark your answers with the answer sheet. Which “I can” statement above still needs work? Look back at the notes for that topic on the site.